

Embedded Linux & System Software Interview Handbook

Phase 6 – Advanced Device Drivers & Hardware Interfaces

This phase focuses on advanced Linux device driver development and the most commonly used hardware interfaces in embedded systems. It bridges the gap between kernel fundamentals and real-world BSP development.

Chapter 13 – Advanced Device Drivers & Hardware Interfaces

Topics Covered

- Linux Device Model Deep Dive
- Platform, I2C, SPI, PCI and USB Drivers
- Character, Block and Network Drivers
- GPIO Subsystem
- UART Driver Internals
- I2C Protocol and Driver Development
- SPI Protocol and Driver Development
- PCIe Architecture
- PCIe Enumeration
- PCI Driver Framework
- USB Architecture
- USB Host and Gadget Drivers
- DMA Fundamentals
- DMA Mapping APIs
- Scatter-Gather DMA
- Interrupt-driven Drivers
- Polling vs Interrupt Mode
- Power Management
- Runtime PM
- Clock Framework
- Reset Controller Framework
- Regulator Framework
- Pinctrl Subsystem
- PWM Subsystem
- RTC Drivers
- Watchdog Drivers
- Sensor Drivers

- Device Tree Integration
- Driver Debugging Techniques
- Real BSP Bring-up Examples
- Production Driver Case Studies
- Senior Interview Questions

Learning Outcomes

After completing this phase, you will be able to design, develop and debug production-quality Linux device drivers, understand hardware communication buses, and perform board bring-up for complex embedded platforms.

Phase	Coverage
Phase 1	Core Linux Kernel Fundamentals
Phase 2	Memory & Process Management
Phase 3	Filesystems & Storage
Phase 4	Linux Networking
Phase 5	Interrupts, Timers & Deferred Work
Phase 6	Advanced Device Drivers & Hardware Interfaces
Estimated Size	300–450 pages when fully expanded
Target	Senior Embedded Linux / BSP / Kernel Interviews